

Counter

1. Design a synchronous counter to count the following arbitrary sequence using JK flip flops and basic gates. [1.5+2.5+2+2]

Sequence : $1 \rightarrow 0 \rightarrow 3 \rightarrow 6 \rightarrow 2 \rightarrow 5 \rightarrow 7 \rightarrow 1 \rightarrow \dots$. Here, the next sequence of the missing number 4 is 6.

- a. Draw the state Diagram
- b. Find the state table with JK flip flop inputs
- c. Minimize the functions of flip flop inputs
- d. Draw the circuit diagram using block Diagram of JK flip flops and basic gates

1. Design a synchronous counter to count the following arbitrary sequence using D flip flops and basic gates. [1.5+2.5+2+2]

Sequence : $0 \rightarrow 2 \rightarrow 1 \rightarrow 4 \rightarrow 3 \rightarrow 7 \rightarrow 6 \rightarrow 0 \rightarrow 2 \rightarrow 1 \dots$, The next number of the missing number 5 will be 0.

- (a) Draw the state Diagram
- (b) Find the state table with D flip flop inputs
- (c) Minimize the functions of flip flop inputs
- (d) Draw the circuit diagram using block Diagram of D flip flops and basic gates

- b) Design a 3 bit asynchronous down counter with positive edge triggered JK Flip flops. [3]

A **MOD-12** counter counts from **0000** to **1011**. Design the MOD-12 binary **ripple** counter with **positive** edge-triggered **T Flip-Flops** and **basic gates**. Include an **asynchronous RESET** signal to reset the circuit to state **0000** (when $RESET = 1$). [

2. Design a 3 bit synchronous up-down counter with two control bits C and D . The circuit counts up when the control variable $C = 1$ and counts down when $C = 0$. Also, the circuit will work if $D = 1$ and remains unchanged if $D = 0$. You can't use more than three negative edge triggered JK Flip flops. Only draw the logic diagram with appropriate labels [5]

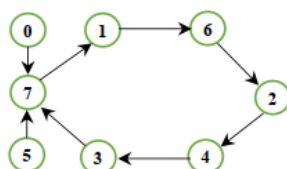
2. Using **J-K flip-flops**, design a synchronous counter that goes through the sequence, 3, 5, 6, 7, 1, 4 ... and repeat. You have to show [5]
 - (a) the flip-flops' input table
 - (b) flip-flops' input equations
 - (c) the logic diagram.

4. Design a 4-bit ripple counter with two control bits C and D . The circuit counts up when the control variable C is logic 'HIGH' and counts down when C is logic 'LOW'. The circuit will work only if $D = 1$ and remains unchanged if $D = 0$. [5]

Q1.	Using J-K flip-flops , design an arbitrary counter that goes through the sequence 2, 4, 3, 7, 1, 0, 5 ... and repeat. You have to show (a) the flip-flops' input table, (b) Simplification of flip-flop's input functions using K-Map and (c) Logic diagram.	(a) JK Flip-Flop [4+4+2]			
		Q (t)	Q (t + 1)	J	K
		0	0	0	X
		0	1	1	X
		1	0	X	1
		1	1	X	0

Q3.	(i) Design a 4 bit asynchronous/ripple up counter with a positive edge triggered JK Flip Flop. You have to include an input named "Count_Up" in your design. The circuit will work if Count_Up = 1 and remain unchanged if Count_Up = 0. (a) Draw the logic diagram of the circuit (b) Explain the design procedure in your own words shortly.	[2+1]
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2. Design an arbitrary synchronous counter circuit using D flip flop that counts according to the sequence given in the given figure. You must show the (i) flip-flop input table, (ii) k-map minimizations of flip-flop input functions and (iii) circuit diagram. [5]



3. A) Design a 4-bit asynchronous Ripple Counter with one control input (A) using negative edge J-K Flip Flops. The counter will count up when A = 1 and will count down when A = 0. [2]

2. Design an arbitrary synchronous counter circuit that counts the sequence given in Figure 1(a). You must show state table, flip-flop input functions, k-map minimizations of output and circuit diagram. You can use any type of flip-flop. [2+2+2+2=8]

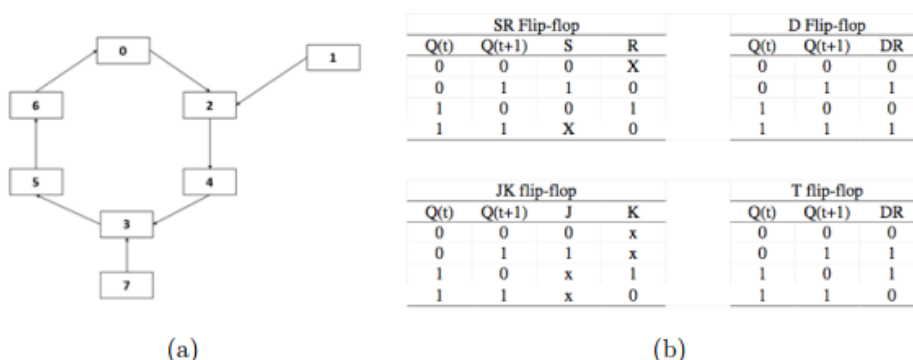


Figure 1: (a)Sequence diagram for Question 2 and (b) Excitation table for different FlipFlops.

1.	Using J-K flip-flops , design a synchronous counter that goes through the sequence 1, 4, 3, 6, 7, 0, 5 ... and repeat. You have to show (a) the flip-flops' input table, (b) equations and (c) the logic diagram.	<table><tr><th colspan="4">(a) JK Flip-Flop</th></tr><tr><th>Q (t)</th><th>Q (t + 1)</th><th>J</th><th>K</th></tr><tr><td>0</td><td>0</td><td>0</td><td>X</td></tr><tr><td>0</td><td>1</td><td>1</td><td>X</td></tr><tr><td>1</td><td>0</td><td>X</td><td>1</td></tr><tr><td>1</td><td>1</td><td>X</td><td>0</td></tr></table>	(a) JK Flip-Flop				Q (t)	Q (t + 1)	J	K	0	0	0	X	0	1	1	X	1	0	X	1	1	1	X	0	[3+ 3+ 2]
(a) JK Flip-Flop																											
Q (t)	Q (t + 1)	J	K																								
0	0	0	X																								
0	1	1	X																								
1	0	X	1																								
1	1	X	0																								
7.	(a) Design a 3-bit ripple up counter using negative edge triggered D-flipflops. You have to draw a neat logic diagram and label all the pins accordingly.		[3+3+2]																								

Sequence Recognizer

2. Design a synchronous sequential circuit to recognize 0010 subsequences including overlaps in an input sequence $x = 0000010010011$, which gives output $y = 0000001001000$,

Design the circuit using T flip flop and basic gates.

[2+2+2+2]

- Draw the state Diagram by assigning state using gray code
- Draw the state table with output and T flip flop inputs
- Minimize the functions of output and flip flop inputs
- Draw the circuit diagram using block Diagram of T flip flops and basic gates

2. Design a synchronous sequential circuit to recognize 1001 subsequences including overlaps in an input sequence $x = 001001001000100100$, which gives output $y = 000001001000000100$, Design the circuit using JK flip flop and basic gates.

[2 + 2 + 2 + 2]

- Draw the state Diagram by assigning state using gray code
- Draw the state table with output and JK flip flop inputs
- Minimize the functions of output and flip flop inputs
- Draw the circuit diagram using block Diagram of JK flip flops and basic gates

1. Design a sequential circuit that can recognize the sequence 1010 using JK flip flop. You have to find the followings:

[8]

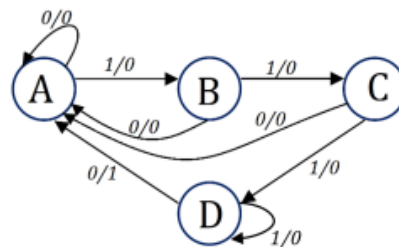
- State diagram.
- State table with Flip-flop inputs.
- K-map minimization of input and outputs.

Design a clocked sequential circuit that recognizes the 4-bit input sequence **1010**, including overlap such that for input $x = 01010100011010010100$ the corresponding output $Z = 0000101000000100001$. Design the circuit using **D Flip-Flops**.

You have to show the following:

- The state diagram
- The flip-flop input table
- The simplified equation(s) for the flip-flop input(s) and the output(s)
- The final circuit diagram

1. The state diagram for recognizing a particular sequence of bits is given below:



i) Identify and write the sequence that's being recognized. Suppose an input bit stream "01110111110" is given to this system. What will be the output stream? [1]

iii) Complete the design of this sequence recognizer using D Flip flops by drawing the state table with Flip-flop's excitation inputs. [3]

(b) Show the K-map minimization of input and output functions. [2]

Q2.	Design a sequential circuit that can recognize the pattern 0010 . Use J-K FF for your design. You must show : (i) State diagram (ii) State table (iii) Flip Flop input functions (iv) K-map minimizations of inputs and output and (v) Circuit diagram.	[2*5=10]
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Answer ALL 3 Questions from Q1 to Q3 [3 × 5 = 15]

- A sequence recognizer is a sequential state machine which takes an input string of bits and generates an output 1 whenever the target sequence has been detected. You have to design a machine where output depends on the present state and the external input (x). Now, design a sequence recognizer for recognizing the sequence 1010 using D flip flop. You have to find the followings. [5]
 - State diagram.
 - Flip-flop input table.
 - K-map minimizations of input and output equations.
 - Logic diagram.
- Design a sequential circuit that can recognize the pattern 1100. Use J-K flip flops for your design. You must show state diagram, state table, flip-flop input functions, k-map minimizations of output and circuit diagram. [2+2+2+2=8]

2.	Using D flip-flops , design a sequence recognizer that can recognize the sequence, “0110”. You have to show (a) state diagram, (b) the flip-flops’ input table, (c) equations and (d) the logic diagram.	[2x4]
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Flip Flop With State Diagram

3. A sequential circuit has one D flip-flop with state Q , two inputs X and Y . Derive the state table and state diagram of the sequential circuit with D as the input to the flip-flop. [8]

$$D = X \oplus Y \oplus Q$$

3. A sequential circuit has two flip-flops (A and B), two inputs (x and y) and an output (z). The flip-flop input functions and the circuit output function are as follows:

$$J_A = xB + y'B', K_A = xy'B', J_B = xA', K_B = xy' + A \text{ and } z = xyA + x'y'B$$

Obtain the logic diagram, state table and the state diagram. [8]

3. a) Describe the undefined or race condition of SR latch with an example circuit.

Design a D Flip-Flop using a JK Flip-Flop and basic gates.

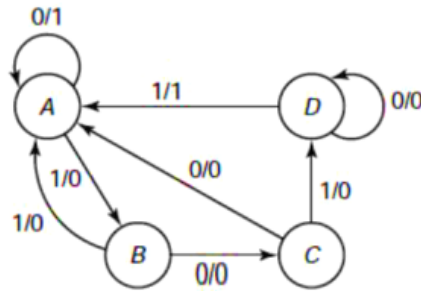
You have to show the following

- The conversion table
- The simplified equation(s) for the flip-flop input(s)
- The final circuit diagram

(b) In a clocked SR NOR latch, why do we use two AND gates in front of the two NOR gates? Give proper reason with a short example. [2]

5. a) Why JK Flip flop is designed in master-slave fashion? Explain your reason properly with an example diagram. [3]

1. Design a sequential circuit for the given state diagram using **D flip-flops**. You must show [5]
 (a) flip-flops' input table
 (b) K-map minimizations of inputs and output
 (c) the logic diagram



Flip-Flop Excitation Tables			
(a) JK Flip-Flop			
$Q(t)$	$Q(t+1)$	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

3. A sequential circuit with two D flip-flops A and B, one input X, and one output Z is specified by the following equations: [5]

$$D_A = \bar{B}X + AX, \quad D_B = \bar{A}X, \quad Z = AB$$
 (a) draw the logic diagram of this circuit
 (b) derive the state table and
 (c) draw the state diagram.
3. A sequential circuit with two D flip-flops (Q is the flipflop output and D is the flipflop input), one input Y, and one output Z is specified by the following input equations:

$$D_1 = Q_2Y + \bar{Q}_1Y, \quad D_2 = \bar{Y}, \quad Z = \bar{Q}_1Q_2$$

For this circuit, i) draw the logic diagram, ii) derive the state table and iv) draw the state diagram. [2+3+3=8]

Register and Flip-flop

4. Design a 4 bit universal shift register which can do the following operations based on the two control INPUTS X and Y as given below. You can use any kind of flip-flops and other necessary gates as required. Draw a neat logic diagram of your solution. [8]

X	Y	Operation
0	0	Left Shift
0	1	Parallel Load
1	0	No Change
1	1	Right Shift

Draw the logic diagram of a 4-bit **universal** shift register, which, based on the two **mode-selection (control) inputs** S_1 and S_0 can do the following operations. You are free to use **any** kind of flip-flop and other necessary gates you prefer.

<i>Mode Control</i>		<i>Register Operation</i>
S_1	S_0	
0	0	Parallel Load
0	1	Shift Left
1	0	Shift Right
1	1	No Change

4. Design a 4 bit register with shift right and shift left operations using D flip flops and basic gates. There Is an asynchronous signal named RIGHT/LEFT, If this bit is 1, the operation will be RIGHT shift, otherwise it will be LEFT shift operation. There will be Serial Right Shift Input (SRI) and Serial Right Shift Output (SRO) for Right shift operation. At the same time there will be Serial Left Shift Input (SLI) and Serial Left Shift Output (SLO) for Left shift operation. The block diagram of the circuit is given in Figure 1(right). [8]

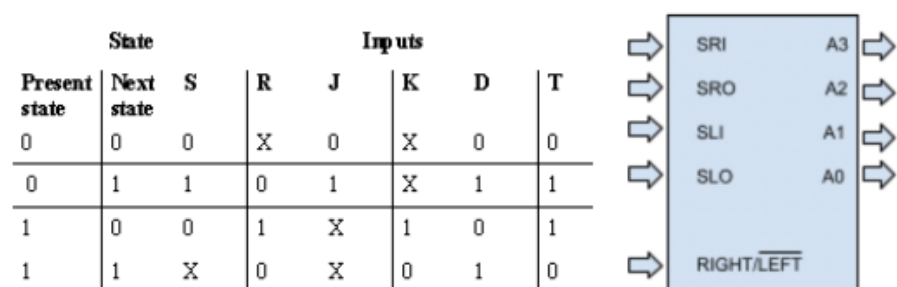
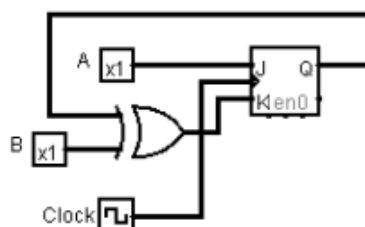


Figure 1: Excitation table for flip-flops (right) and block diagram for ripple register (right).

5. a) You are analyzing the output of the following circuit with one JK Flip flop and two inputs A and B. Assume the initial output Q = 1. [3]



What will be value of Q and name of the operation(Set/Reset/Complement/Unchanged) if you give the following three inputs to the circuit one after another and why?

- (i) Clock = 1, A = 1 and B = 0
- (ii) Clock = 0, A = 1 and B = 0
- (iii) Clock = 1, A = 1 and B = 0

- B) We have a lot of D flip flops at our disposal. Design a 4-bit register using as many D flip flops along with any number of basic gates or combinational functional blocks as needed that can do the following three operations: load, no change and reset. [3]

4. Design a 4-bit register that can perform shift right or shift left based on a control bit C. If C=0, the register will shift right and if the control bit C=1, the register will shift left. [8]

6.	A sequential circuit with two D flip-flops A and B, two inputs X and Y, and one output Z is specified by the following input equations: $D_A = BY + \bar{A}Y, D_B = \bar{Y}, Z = \bar{A}\bar{B}$ (a) Draw the logic diagram of the circuit. (b) Derive the state table. (c) Derive the state diagram.	[3+2+2]
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(b) Design a 4-bit shift left register using any type of flip-flop. You have to draw a neat logic diagram and label all the pins accordingly.

Priority Encoder + Encoder

5. Consider an application of magnetic positional control for ship navigation. There are six possible events here: North (NO-000), North-East (NE-001), East (EA-011), South (SO-110), South-West (SW-010), West (WE-101). Here, the first two letters are the input and the three bits are the output. For example, in NO-000, NO is the input and 000 is the output. To maintain priority during the occurrence of multiple events at the same time, the priority sequence followed is
NO > NE > SW > WE > EA > SO. Design an 6 X 3 encoder that can encode the events based on given priority. You have to answer the followings: [2+3+3]
- Derive the truth table of the priority encoder including the valid bit.
 - Derive the Boolean expressions for all the outputs.
 - Draw the logic diagram using basic gates.
5. Consider a UIU alarm system with five inputs: fire (F-011), water-tank-full (W-001), burglary (B-100), smoke (S-111) and dust (D-010). For this alarm system an alarm signal (A) is on if any of these events occur a three bit code is shown to denote the incident. In case of multiple events, a priority is followed as: $F > B > S > D > W$. For this alarm system, find out the simplified expression of the alarm signal (A) and the encoding bits for the associated 5×3 priority encoder and draw the logic diagram. [8]
2. a) Consider a system which indicates which part of a laptop is damaged. There are six possible events here: fan damaged (F-001), keyboard damaged (K-111), speaker damaged (S-010), track-pad damaged (T-101), ethernet port damaged (E-011) and USB port damaged (U-100). Here, the first letter is the input and the three bits are the output. For example, in **F-001**, F is the input and **001** is the output. To maintain priority during the occurrence of multiple events at the same time, the priority sequence followed is $F > K > T > E > U > S$. Design an encoder that can encode the events based on given priority. You have to answer the followings: [5]
- Derive the truth table of the priority encoder including valid bit.
 - Derive the Boolean expressions for all the outputs.

7.	A) Design a 4-input Priority Encoder with inputs $D_3D_2D_1D_0$. Derive the truth table of the priority encoder where priority is set according to the following sequence of priority from the highest to the lowest $D_1 > D_2 > D_0 > D_3$. Derive the Boolean expressions for outputs and draw the circuit diagram.	
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4. a) Design a decimal to excess-3 encoder which can only take 3 bits of input. Derive the truth table and the boolean expression for all the outputs. No need to draw logic diagram. [3]

Answer any 2 Questions from Q5 to Q7 [2×5=10]

5. Design a 4-input Priority Encoder with inputs D_0, D_1, D_2, D_3 and outputs Y_1, Y_0 . Y_1 and Y_0 are the two bits representing the binary equivalent of the inputs D s. For example, D_0 is encoded to 00, D_1 to 01 and so on. [5]
- Derive the truth table of the priority encoder where priority is set according to the following sequence of priority from highest to lowest $D_1 > D_3 > D_0 > D_2$.
 - Derive the Boolean expressions for Y_1 and Y_0 .
 - Draw the circuit diagram.

(ii) Consider a UIU alarm system with five inputs: fire (F-001), water-tank-full (W-101), burglary (B-100), smoke (S-111) and dust(D-000). For this alarm system an alarm signal (A) is on if any of these events occur and the corresponding three bit output code is shown to denote the incident. For this alarm system:	[1+2+1]
- Find out the simplified expression of the alarm signal (A) - The design of the encoder system(Output Equations) for the given events. - Draw the logic diagram.	

6. Consider an UIU alarm system with five inputs: fire (F-001), water-tank-full (W-101), burglary (B-100), smoke (S-111) and dust (D-000). Here, the first letter is the input and the three bits are the output. For example, in "F-001", F is the input and "001" is the output. To maintain priority during the occurrence of multiple events at the same time, the priority sequence followed is $D < W < B < S < F$. [5]
- Derive the truth table of the priority encoder.
 - Derive the Boolean expressions for all the outputs.

5. Design a 4-input Priority Encoder with inputs D_0, D_1, D_2, D_3 and outputs Y_1, Y_0 . Y_1 and Y_0 are the two bits representing the binary equivalent of the inputs D s. For example D_0 is encoded to 00, D_1 to 01 and so on. Derive the truth table of the priority encoder where priority is set according to the following sequence of priority from highest to lowest $D_0 > D_1 > D_3 > D_2$. Derive the Boolean expressions for Y_1 and Y_2 and draw the circuit diagram. [8]

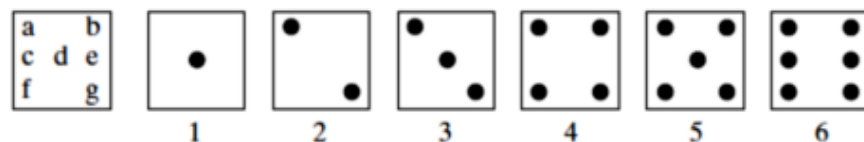
5.	Consider a UIU alarm system with five inputs: fire (F-011), water-tank-full (W-001), burglary (B-100), smoke (S-111) and dust(D-010). For this alarm system an alarm signal (A) is on if any of these events occur a three bit code is shown to denote the incident. In case of multiple events, a priority is followed as: F>B>S>D>W. For this alarm system, find out the simplified expression of the alarm signal (A) and the encoding bits for the associated 5x3 priority encoder and draw the logic diagram.	[8]
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Decoder

6. Answer all the following questions below. [6+2]

- a. An electronic game uses an array of seven LEDs (light-emitting diodes) to display the results of a random roll of a die. Use a 3:8 active low line decoder and OR gates to map the 3-bit combinations on inputs X2, X1, and X0 for values 1 through 6 to the outputs a through g. Input combinations 000 and 111 are don't-cares.



7. (a) Implement a 3-to-8 decoder using two 2-to-4 decoders and a single NOT gate (you can not use any other gates). Label the decoder pin names and their inputs/outputs carefully. [4]

- b) Implement $F(A, B, C) = \Pi_M(0, 1, 5, 6, 7)$ using 2-to-4 line active high decoders and basic gates. Draw the logic diagram with proper symbols. [5]

A) Design the following Boolean function using a 3-to-8 line decoder and an OR gate.

$$F(A, B, C) = A\bar{B} + B\bar{C}$$

b) Implement the following two functions using 2-to-4 decoders.

$$F(A, B) = \Pi_M(0, 1)$$

$$F(X, Y) = \sum_m(0, 1)$$

(i) Draw the logic diagram with proper symbols.

(ii) Can you complete the design with a single decoder? Why or why not?

b) Implement a 3×8 line decoder using 1×2 line decoders only.

(b) Suppose you have to build a circuit with a 4-bit number as input and 2 output bits, O_1 and O_0 . [3]

- If the input number is even, O_1 will be '1' otherwise it will be '0'.
- If the input number is divisible by 6, O_0 will be '1' otherwise it will be '0'.

Implement this circuit using a single decoder of your choice and OR gates only.

	(iii) Draw the implementation of the following functions using active high decoders. $F1(A, B, C) = \sum m(0, 2, 4, 5)$ $F2(X, Y, Z) = \prod M(1, 2, 6, 7)$	[3]
	(ii) Using any number of 3 to 8 active high decoders, implement a 5 to 32 active high decoder. You must draw the circuit diagram for your design and label the decoder pin names and their inputs/outputs carefully.	[3]

4. - You have to build a circuit with 3-bits number (A, B, C) as input and 2-bits number (X, Y) as output. [5]

- If the total number of ones in the input bit is even then X will be '1' else X will be '0'. For example, if input is "101" then the X will be '1'.
- If the decimal equivalent of the input is odd then Y will be '1' else Y will be '0'. For example, if the input "101" then Y will be '1'.

Implement the circuit using a single decoder of your choice and OR gates only.

(b) Implement a 2-to-4 line decoder using 1-to-2 line decoders only.

- (b) A majority function has an output value of 1 if there are more 1s than 0s on its inputs. The output is 0 otherwise. On the other hand, a parity function gives output 0 if there are even number of 1s and output 1 if there are odd number of 1s. You have to implement both of the functions using a single decoder and OR gates only. Consider 3-bit input. [4]

Section B: Answer any two of the Questions from 3, 4 and 5.

3.	(a) Implement a 3-to-8 decoder using two 2-to-4 decoders and a single NOT gate. Label the decoder pin names and their inputs/outputs carefully.	[4+4]
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Multiplexer

- b. Design a 16:1 multiplexer with 2:1 multiplexers. You can use any number of 2:1 multiplexers. You need to use the block diagram of the 2:1 multiplexer.
- a. Implement the following Boolean function using a 4:1 MUX and necessary basic gates.
- $$F(A, B, C, D) = \sum m(0, 1, 3, 4, 6, 7, 10, 12, 13, 14, 15)$$
6. (a) Design a 16×1 multiplexer using 4×1 multiplexers only. Draw the logic diagram and show the working principle with an example. [4]
- (b) A combinational circuit is defined by the following Boolean function:

$$F = \overline{x + z} + xyz$$

Design the circuit with a 2×1 MUX and a decoder (you can not use any other gates). [4]

- (b) Consider a digital system with four binary bits as input and a single bit as output. The binary output is 1 if the number of 1s in the input is less than the number of 0s, otherwise the output is 0. Implement this system using a 8×1 MUX and a single NOT gate only (you can not use any other gates). [4]

b) Implement $F(A, B, C) = \Pi_M(1, 3, 5, 6)$ using one 2-to-1 line mux and logic gates. Draw truth table and logic diagram. [5]

4. a) Design a logic function that will take a 4 bit binary number as input and output if the total number of 1 in input is greater than 2 or not. For example, if input is 0011, then output is 0, if input is 0111, output is 1. Now implement this function with 4x1 MUX. You can use more than one 4x1 MUX but you can't use any basic gates(AND,OR,XOR,NOT etc).

(i) Draw the truth table [4]

(ii) Draw the logic diagram [4]

6. A) Implement the following Boolean function with an 8-to-1-line multiplexer and external basic gates.

$$F(W, X, Y, Z) = \Pi_M(0, 2, 5, 6, 7, 8, 9, 10)$$

B) Design an 8-to-1 -line multiplexer using 2-to-1 line multiplexers only. You need to use the

block diagram of the 2 to 1 line multiplexer.

3. Design a function that will take a 3 bit binary number as input and the output will be "HIGH" if total number of 1s in input is less than or equal to total number of 0s and "LOW" otherwise. Now implement this function with 2x1 MUX. You can use more than one 2x1 MUX if necessary but you can't use any basic gates(AND,OR,XOR,NOT etc).

(i) Draw the truth table [2]

(ii) Draw the logic diagram [2]

6. a) Is it possible to use a 16×1 MUX as a 4×1 MUX? Explain your answer with necessary diagram. [2]

7. (a) Implement the following Boolean function using a 8×1 line multiplexer and other basic gates. [2]

$$F(A, B, C, D) = \prod M(0, 2, 3, 7, 9, 13, 14, 15)$$

number of 2-bit full adders as needed.

B) Implement the function $F(A, B, C, D) = \sum(0,3,6,8,12,14,15)$ using a 4x1 MUX.

6. (a) Implement the following Boolean function using a 4-to-1 line multiplexer and other necessary gates. [5]

$$F(A, B, C, D) = \prod M(1, 2, 4, 7, 8, 9, 12, 13, 14, 15)$$

7. (a) Design a 16:1 MUX using 4:1 MUX (as many as you require) only.

(b) Consider a digital system with four binary bits as input and a single bit as output. The binary output is 1 if the number of 1s in the input is less than the number of 0s, otherwise the output is 0. Implement this system using a 8x1 MUX and a single NOT gate only.

4. (a) Design a 16x1 multiplexer using 4x1 multiplexers only. Draw the logic diagram and show the working principle with an example. [4+4]

Parallel Adder

- b. Draw the diagram of a parallel adder to perform addition of the numbers $(9)_{10}$ and $(2)_{10}$. First, you need to determine the size of the required parallel adder. After that, you need to draw the diagram and show the process to add the two numbers that are given.

TABLE 1-3 Excitation Table for Four Flip-Flops

SR flip-flop				D flip-flop		
$Q(t)$	$Q(t + 1)$	S	R	$Q(t)$	$Q(t + 1)$	D
0	0	0	×	0	0	0
0	1	1	0	0	1	1
1	0	0	1	1	0	0
1	1	×	0	1	1	1

JK flip-flop				T flip-flop		
$Q(t)$	$Q(t + 1)$	J	K	$Q(t)$	$Q(t + 1)$	T
0	0	0	×	0	0	0
0	1	1	×	0	1	1
1	0	×	1	1	0	1
1	1	×	0	1	1	0